

(Time: 2 hours)

[Total Marks: 60]

- N. B.: (1) **All** questions are **compulsory**.
 (2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.
 (3) Answers to the **same question** must be **written together**.
 (4) Numbers to the **right** indicate **marks**.
 (5) Draw **neat labeled diagrams** wherever **necessary**.
 (6) Use of **Non-programmable** calculator is **allowed**.

1. **Attempt any two of the following:** 12
 - a. In the DevOps context, what do you improve? Explain why the heart of DevOps lies in achieving continuous improvement.
 - b. How can organizations effectively embrace DevOps practices to drive innovation and agility in the rapidly evolving application economy?
 - c. State how monolithic and microservice application architectures differ, and what operational challenges are associated with monolithic designs that microservices can address to offer significant business benefits.
 - d. Give some examples of metric classes that can have adverse effects on DevOps practices. How do these metrics undermine the principles of effective DevOps implementation?

2. **Attempt any two of the following:** 12
 - a. Describe how API performance monitoring extends the functionality of API management and Application Performance Management (APM) solutions and explain its essential integrations for effective API management.
 - b. To address the complex issues involved with acquiring quality data and ensuring regulatory compliance, what features and capabilities should modern test data management solutions provide?
 - c. Design a capability checklist for a release automation solution to ensure it delivers a deployment engine capable of supporting essential deployment requirements.
 - d. What are major considerations the organizations need to address in order to ensure they have the right application performance management strategies and tooling for delivering agile operations and shift-left approaches?

3. **Attempt any two of the following:** 12
 - a. What are the two critical areas that any team developing and executing a DevOps transformation roadmap needs to understand and prepare for?
 - b. How can techniques be leveraged to determine business value numbers in the context of a DevOps transformation?
 - c. Find out how channels function as the various methods through which a line of business (LOB) and an IT organization deliver their value propositions to different customer segments. What is their significance in ensuring effective value delivery?
 - d. Summarize the key DevOps plays for establishing metrics and KPIs to optimize the delivery pipeline and explain how these metrics and KPIs contribute to enhancing the efficiency and effectiveness of the pipeline.

4. **Attempt any two of the following:** 12
- a. How can leveraging existing reference sources, roles and practices enhance practical DevOps, particularly in relation to DevOps and IT Service Management?
 - b. What does a real-world model for DevOps ROI look like? How can it be effectively applied to evaluate the return on investment for DevOps initiatives?
 - c. Discuss the key considerations for talent management in DevOps. How can organizations ensure that they have the right skills and mindset to effectively implement DevOps practices?
 - d. Find out the role of 'Enabling Experimentation' as a core theme in driving innovation. How it supports the development and validation of new ideas within an organization.
5. **Attempt any two of the following:** 12
- a. How does developing a culture of innovation at scale contribute to scaling DevOps in an enterprise? What strategies can organizations use to foster this culture effectively?
 - b. What are the key security considerations for scaling DevOps within an enterprise? How can organizations effectively integrate security practices into their DevOps processes to protect their systems and data?
 - c. Discuss the key strategies for leading DevOps adoption in the enterprise. How leaders can effectively guide their organizations through the transition to DevOps practices.
 - d. Why does focusing on people play a crucial role in leading DevOps adoption within the enterprise? What are the key strategies for effectively managing and supporting individuals through this transformation?
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